

COSC 1336 – Fundamentals of Programming Project Assignment #6 - Grade Statistics

Overview

Write a Python program that reads a file named data.txt, which contains a list of integer values (e.g., student grades). The program will read the file, create a list of the numbers, and compute several statistics.

File Format (data.txt):

Each line contains a single integer.

```
90
100
80
```

Program Requirements

Using list methods and functions, the program should:

- Display all the numbers in the list
- Display the **count** of numbers
- Display the **sum** of the numbers
- Calculate and display the **average**
- Display the **maximum** value
- Display the **minimum** value

Function Requirements

Your program **must use functions**. At a minimum, include the following:

- readData() – Reads the file and returns a list of integers
- main() – Manages the main logic of the program

Optional

You may create additional functions to improve code clarity and organization.

Additional Notes:

- Validate that the file exists and handle errors gracefully.
- Use a professional programming style: include comments, meaningful variable names, and consistent formatting.

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Sample Output

Project #6

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Grade Statistics Summary

The numbers are 90, 100, 80

The count of numbers is 3

The sum of numbers is 270

The average of the numbers is 90.00

The max number is 100

The min number is 80

End of Project 6